Bayesian Discrete-Time Survival Modeling via Pólya-Gamma Augmentation: An Application to ICU Mortality

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1. Introduction

Accurate mortality risk profiling in the Intensive Care Unit (ICU) is essential for clinical decision-making, resource allocation, and the design of adaptive treatment strategies. Standard continuous-time approaches such as the Cox Proportional Hazards model remain dominant in biomedical survival analysis. However, when applied to large-scale electronic health records such as MIMIC-III (Johnson et al., 2016), handling of censoring is often treated as an unexamined formality. Cox and discrete-time models typically rely on the non-informative censoring assumption meaning that the censoring mechanism (e.g., hospital discharge) carries no information about the underlying survival process beyond what is captured by observed covariates (Kalbfleisch and Prentice, 2003).

In the ICU context, this assumption is immediately suspect. A patient that is "censored" is the result of a purposeful clinical decision. This clinically determined process is likely to be systematically related to the patient's underlying health status and mortality risk. Thus, this dependence between the censoring C_i and and unobserved failure process T_i violates the standard and biases downstream inference of hazard estimates, coefficient effects, and predicted survival functions.

The consequences of informative censoring have been studied in the missing data literature. Rubin (1976) formalized the missingness taxonomy, that maps directly to censoring. The main interest here is the consequences of violations to the Missing at Random (MAR) assumption, which corresponds to the non-informative censoring assumption. Further, Tsiatis (1975) shows a fundamental identifiability issue; the observed data (y_i, δ_i) alone, the marginal distribution of T_i and C_i cannot be identified without stable structural assumptions on their joint distribution. This means that diagnostics based on the observed data alone cannot distinguish informative from non-informative censoring. Any asessment of the censoring mechanism but rely on untestable structural assumptions, thus motivating the need for sensitivity analyses in Section 4.3.

The central question of this project is: *under what conditions does the independent right-censoring assumption hold in ICU survival data, and what are the consequences for Bayesian posterior inference when it is violated?* We address this question through three methodological contributions, all developed within a discrete-time Bayesian framework using the person-period formulation of Allison (1982).

First, we formalize the independent censoring assumption within the context of missing data, derive standard discrete-time likelihood factorization, and characterize the consequences of violations to this assumption (Section 3). Second, we propose a *shared frailty model* that introduces a patient-level random effect b_i to capture unobserved heterogeneity that confounds both the censoring decision and the mortality process (Section 4). Third, we will implement a *exponential tilting sensitivity analysis* following the selection model approach of (Scharfstein et al., 1999) which introduces a fixed sensitivity parameter γ that indexes the log-odds ratios of mortality between patients who are censored and those who remain under observation (Section 4.3). By finding the γ^*, the tipping point at which critical clinical conclusions change, this will provide a principled way to evaluate the robustness of the frailty model to informative censoring. Finally, we will discuss other extensions, such as competing risks and how this might be fruitful for future work (Section 6). The computational backbone for all models is the Pólya-Gamma data augmentation scheme (Polson et al., 2013), which allows for efficient Gibbs sampling of the regression coefficients and frailty terms in a conjugate manner (Section 5). This sidesteps the computational challenges of high-dimensional MCMC and Metropolis-Hastings tuning for logistic hazard models.

2. Data Structure, Person-Period Format, & Notation

Let $i = 1, \ldots, N$ index unique ICU admissions from MIMIC-III. We define:

- $T_i > 0$: the latent time to in-hospital death (continuous, in days).
- $C_i > 0$: the censoring time, defined as the time of hospital discharge if the patient does not die, i.e., the time at which the patient leaves the study for any reason, alive. In MIMIC-III, this is a clinically determined event, not an administrative one.
- $y_i = \min(T_i, C_i)$: the observed follow-up time (days).
- $\delta_i = \mathbb{I}(T_i \leq C_i)$: failure indicator.
- $\mathbf{x}_i \in \mathbb{R}^P$: a vector of static baseline covariates (age, admission type, comorbidity scores, etc.), measured at time-zero (ICU admission).

Discretization. We partition continuous time into discrete intervals of width Δ: The intervals are, $[0, \Delta), [\Delta, 2\Delta), \ldots, [(K-1)\Delta, K\Delta)$ with $t = 1, \ldots, K$ indexing those intervals. The discrete time failure time is $T_i = \lceil T_i/\Delta \rceil$ and similarly the discrete censoring time is $C_i = \lceil C_i/\Delta \rceil$. Our primary analysis will use week-length intervals ($\Delta = 7$ days), yielding $K \approx 4 - 5$ intervals. To assess sensitivity to this choice, we will also explore bi-weekly intervals ($\Delta = 14$ days, $K = 5$) and day-length intervals ($\Delta = 1$ day, $K = 70$) in Section 4.3. The finer the discretization, the better the approximation to the underlying continuous-time process, but the larger the dataset and more parameters to estimate.

Person-Period expansion. The data is restructured into a "person-period" format (Allison, 1982), so that patient i contributes one row per interval $t = 1, \ldots, \tilde{y}_i$, yielding the binary outcome:

$$y_{i,t} = \mathbb{I}\left(T_i \in [(t-1)\Delta, t\Delta), \delta_i = 1\right) = \begin{cases} 1 & \text{if patient } i \text{ dies in interval } t \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Censored pateints contribute $y_{i,t} = 0$ for $t = 1, \ldots, \tilde{C}_i$ and are removed from the risk set thereafter. Patients who die contribute $y_{i,t} = 1$ for $t = 1, \ldots, \tilde{T}_i$ and exit. The total number of persion-period rows is $\mathcal{R} = \sum_{i=1}^{N} \tilde{y}_i$.

3. Standard Likelihood & Censoring Assumption

Here, I will introduce the standard likelihood and assess the direct impacts under violations of the independent censoring assumption. This will motivate the need for the frailty model and sensitivity analysis extensions in Sections 4–4.3.

In (K&P, Section 2.2), the discrete-time Bernoulli likelihood for patient i is:

$$h_{i,t} = \Pr(Y_{i,t} = 1 \mid Y_{i,t-1} = 0, \mathbf{x}_i) = \Pr(T_i = t \mid T_i \geq t, \mathbf{x}_i), \quad t = 1, \ldots, K_i \tag{2}$$

for K_i being the last interval for which patient i is at risk (i.e., $K_i = \tilde{y}_i$). The corresponding survival function is:

$$S_i(t) = \Pr(T_i > t \mid \mathbf{x}_i) = \prod_{s=1}^{t} (1 - h_{i,s}) \tag{3}$$

3.1 Full Likelihood.

Given right-censoring, the likelihood contribution for patient i, involves the joint density of (y_i, δ_i) given $\mathbf{x}_i$, which in principle depends on the failure (parameters $\boldsymbol{\theta}$) and censoring processes (parameters $\boldsymbol{\phi}$). Let f_T, S_T, and f_C, S_C denote the densities and survival functions of T_i and C_i. The full data likelihood is (K&P, Section 3.2):

$$\mathcal{L}_i^{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \big[f_T(y_i \mid \mathbf{x}_i)\, S_C(y_i \mid \mathbf{x}_i)\big]^{\delta_i} \big[S_T(y_i \mid \mathbf{x}_i)\, f_C(y_i \mid \mathbf{x}_i)\big]^{1-\delta_i}, \tag{4}$$

This expression cannot be seperated into independent parts of $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ unless failure and censoring processes are conditionally idependent (i.e. $T_i \perp C_i \mid \mathbf{x}_i$). This non-separability reflects the fundamental identifiability problem established by Tsiatis (1975): fron infinitely many joint distributions of (T_i, C_i) that are consistent with the observed data (y_i, δ_i).

3.2 Independent Censoring Assumption

Assumption: *[Independent Right-Censoring; K&P Section 2.4] The censoring mechanism is* non-informative *in the sense that:*

$$T_i \perp C_i \;\mid\; \mathbf{x}_i. \tag{5}$$

Equivalently, at any time t in which patient i is under observation, knowing the exact censoring time C_i provides no additional information about future survival beyond what $\mathbf{x}_i$ already encodes.

Under Assumption 3.2, the full likelihood (4) factors as:

$$\mathcal{L}_i^{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \underbrace{[f_T(y_i \mid \mathbf{x}_i)]^{\delta_i}\, [S_T(y_i \mid \mathbf{x}_i)]^{1-\delta_i}}_{\mathcal{L}_i(\boldsymbol{\theta})} \quad \underbrace{[S_C(y_i \mid \mathbf{x}_i)]^{\delta_i}\, [f_C(y_i \mid \mathbf{x}_i)]^{1-\delta_i}}_{\text{depends only on } \boldsymbol{\phi} \text{ (under Assumption 3.2)}} \tag{6}$$

so that inference about $\boldsymbol{\theta}$ can be processed using the observed-data likelihood $\mathcal{L}_i(\boldsymbol{\theta})$ without any reference to the censoring mechanism (K&P, Equation 3.1). In discrete-time person-period form:

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{N} \prod_{t=1}^{K_i} h_{i,t}^{y_{i,t}} (1 - h_{i,t})^{1-y_{i,t}}$$

where each (i, t) term is treated as an independent Bernoulli trial.

3.3 Censoring Condition & Operative Assumption

Independent censoring (Assumption 3.2) corresponds to Missing at Random (MAR) in Rubin (1976)'s taxonomy of missing data. This means that conditional on $\mathbf{x}_i$, the censoring mechanism carries no information about T_i. Since $\mathbf{x}_i$ contains only baseline measurements and MIMIC-III discharge decisions, are likely more reprepresentative using time-varying covarites, the MAR assumption is likely violated in the ICU context, and thus the censoring mechanism is likely informative to some degree. This motivates the operative censoring condition (Assumption 3.3), which is the working assumption that the MAR assumption is violated, and thus we will proceed with methods that account for the degree of informative censoring, such as the frailty model (Section 4) and the sensitivity analysis (Section 4.3).

Assumption: *[Operative Censoring Condition] In our static covariate setting, MAR given only $\mathbf{x}_i$ is the operative censoring requirement. Because time-varying information is likely critical to drive discharge decisions, the censoring mechanism is likely informative (i.e., $T_i \not\perp C_i \mid \mathbf{x}_i$). Thus, we will proceed under the assumption that the MAR assumption is violated, and work towards developing methods to account for this.*

3.4 Consequences of Informative Censoring

When the independent censoring assumption fails, three consequences follow.

Biased Hazard Estimation. First, if the healthier patients are systematically discharged before sicker ones, the risk set at time t is biased towards sicker patients, inflating the observed hazard $\hat{h}_t$ relative to the true marginal hazard h_t^*. The risk set at time t is $\mathcal{R}_t = \{i : y_i \geq t\} = \{i : T_i \geq t, C_i \geq t\}$ and $\mathcal{C}_t = \{i : C_i < t\}$. The bias is (Appendix B):

$$\hat{h}_t \approx h_t^* + \frac{\lambda_t \Delta_t}{1 - \lambda_t} \quad \text{where} \quad \hat{h}_t = \frac{\sum_{i \in \mathcal{R}_t} \mathbb{I}(T_i = t)}{|\mathcal{R}_t|} \tag{7}$$

where $\lambda_t = |\mathcal{C}_t|/|\mathcal{R}_t|$ is the proportion of surviving patients censored by time t, and $\Delta_t > 0$ is the hazard difference between the full survivor population and the censored subgroup. The bias is positive for all $\lambda_t, \Delta_t > 0$ and increases as $\lambda_t \to 1$ (i.e. more patients are discharged over time, shrinking the risk set), and is the discrete-time analogue of the frailty-induced selection bias in the continuous-time setting (Vaupel et al., 1979; Lancaster, 1979). See Appendix B for a full derivation of the bias in the observed hazard under informative censoring.

Since the observed hazard being positively biased in each interval, this implies that the estimated survival function $\hat{S}(t) = \prod_{s=1}^{t}(1 - \hat{h}_s)$ will compound this bias and be systematically lower than the true marginal survival $S^*(t)$. This means that patients with long ICU

stays (e.g., 10+ days) will have their survival estimates show a large discrepancy from the true marginal survival. This leads to a pessimistic empirical estimate of survival.

Confounding Bias in Covariate Effects. Covariate information that might predict early discharge and lower mortality risk (e.g., elective admission, lower comorbidity scores) will have their effects $\boldsymbol{\beta}$ confounded by the informative censoring process. Thus, estimand derived in the standard likelihood will no longer represent the pure effect of $\mathbf{x}_i$ on mortality risk, but rather both the effect of $\mathbf{x}_i$ on the failure and censoring processes. Scharfstein et al. (1999) formalizes this as a failure of the semiparametric estimation equations when MAR is violated, where the score function for $\boldsymbol{\beta}$ is biased proportional to the dependence between T_i and C_i given $\mathbf{x}_i$. This is formalized in Appendix A, where the bias in the score function for $\boldsymbol{\beta}$ under informative censoring can be expressed as:

$$\text{Bias}(\alpha) \approx -\alpha\, \mathbb{E}\Big[\Lambda_0(C_i \mid \mathbf{x}_i)\, T_i\, S_{\text{full}}(\boldsymbol{\beta})\Big] \tag{8}$$

In the context of this project, the full-data score ($S_{\text{full}}(\boldsymbol{\beta})$) is Equation 4. This bias is driven by α (the severity of the MAR violation/selection) and the covariance (cumulative hazard) between the failure time T_i and the censoring $\Lambda_0(C_i \mid \mathbf{x}_i)$. Because α cannot be identified from the observed data alone, standard estimation fails.

Corrections. The consequences yield a upward systematic bias in observed hazards and a multiplicative bias in the score function for $\boldsymbol{\beta}$, depending on severity of the MAR violation. First, the shared frailty model (Section 4) partially restores the risk-set representativeness by conditioning on unobserved heterogeneity b_i, and the sensitivity analysis (Section 4.3) directly parameterizes the residual bias through γ, allowing us to assess how large the MAR violation-effect would need to be before changing clinical conclusions.

4. Logistic Hazard Model & Frailty Extension

The persion-period formulation in Section 2 recasts the discrete-time survival hazard as a first-order Markov chain on the state space $\{1, 0\}$ where 0 denotes "alive" and 1 denotes "dead" (an absorbing state). At each interval t, a patient who survives to that point ($Y_{i,t-1} = 0$) either transitons to death ($Y_{i,t} = 1$) or remains alive ($Y_{i,t} = 0$). Note that conditioning on $Y_{i,t-1} = 0$, also conditions on all previous time intervals while also satisfying the Markov property. The discrete-time hazard (2) is precisely the one-step transition probability, which can be modeled as a Bernoulli GLMM with a logistic link.

4.1 Hierarchy Specification

We model the hazard via a logistic link augmented with patient-level Gaussian random effect (frailty) b_i:

$$Y_{i,t} \mid h_{i,t} \sim \text{Bernoulli}(h_{i,t}), \quad t = 1, \ldots, K_i,\ i = 1, \ldots, N, \tag{9}$$

$$\text{logit}(h_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + b_i, \tag{10}$$

with priors:

$$\alpha_1 \sim \mathcal{N}(0, \sigma_0^2), \qquad \text{(anchor)} \tag{11}$$

$$\alpha_t \mid \alpha_{t-1} \sim \mathcal{N}(\alpha_{t-1}, \sigma_\alpha^2), \qquad t = 2, \ldots, K, \tag{12}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_P), \tag{13}$$

$$b_i \sim \mathcal{N}(0, \sigma_b^2), \qquad i = 1, \ldots, N, \tag{14}$$

$$\sigma_b^2 \sim \text{Inv-Gamma}(a_0, b_0), \tag{15}$$

$$\sigma_\alpha^2 \sim \text{Inv-Gamma}(a_\alpha, b_\alpha) \tag{16}$$

The first-order random walk prior on $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)^\top$ endosed the belief that neighboring intervals have similar baseline log-odds of death. In matrix form, the joint prior ha precision $\boldsymbol{\Sigma}_\alpha^{-1} = \sigma_\alpha^{-2} \mathbf{D}^\top \mathbf{D}$ where $\mathbf{D}$ is the $(K-1) \times K$ first-difference matrix. As the risk set shrinks at later intervals, this prior borrows strength from adjacent time intervals, regularizing α_t where data is sparse. The Inverse-Gamma (IG) hyperpriors are chosen for their conjugacy properties, allowing the Pólya-Gamma augmentation scheme. We set $a_0 = b_0 = a_\alpha = b_\alpha = 0.01$ and $\sigma_0^2 = \tau^2 = 10$ as weakly informative defaults.

Let $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathbb{R}^{K+P}$ denote the combined fixed-effect vector, with joint prior $\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \Sigma_0)$ such that $\Sigma_0 = \text{blockdiag}(\Sigma_\alpha, \tau^2 \mathbf{I}_P)$ is a block-diagonal covariance matrix reflecting the temporal smoothing for $\boldsymbol{\alpha}$ and independent regularization for $\boldsymbol{\beta}$. Setting $b_i = 0$ recovers the standard logistic hazard model, so comparing the two through posterior summaries of σ_b^2 provides a direct diagnostic for unmeasured heterogeneity.

4.2 Why Frailty Addresses Informative Censoring

The frailty term b_i captures the heterogeneity which may drive both discharge decisions and mortality risk. By codnitioning on the random effect, the model weakens the censoring requirement (Assumption 3.2) to:

$$T_i \perp C_i \mid \mathbf{x}_i, b_i \tag{17}$$

This strictly weaker condition, requires only that the augmeted covariate set $(\mathbf{x}_i, b_i)$ renders censoring non-informative (Daniels and Hogan, 2008, Chapter 7). Under (17), the conditional analogue of the bias term in (7) vanishes, because both censored and uncensored subpopulations have the same conditional hazard distribution given $\mathbf{x}_i, b_i$.

An important caveat is that the bias formula in (7) assumes $\Delta_t > 0$, meaning censored patients are healthier. In the ICU context, this is likely an oversimplification. While the majority of patients are likely to improve $b_i < 0$, those who are discharged to hospice for example, might have a worse prognosis $b_i > 0$. For these patients, $h_t^c > h_t^*$ locally, attenuating or even reversing the bias. The frailty model handles this naturally by estimating b_i individually. But whether (17) holds is ultimately untestable, which is why the sensitivity analysis in Section 4.3 remains essential.

Identifiability. The baseline hazard α_t and frailties b_i are seperately identified by the temporal structure of the data. The random walk α_t varies across intervals and b_i bu correlation across intervals within patient i. However, patients observed for only one interval

($K_i = 1$) contribute no within-patient information and their posterior is entirely determined by the prior, shrinking b_i towards zero.

The practical implication of this is that the conjugate full conditional (derived in Section 5), the large number of prior-dominated frailties ($b_i \approx 0$ for $K_i = 1$ patients) pull $\hat{\sigma}_b^2$ downwards, potentially underestimating heterogeneity among patients with longer stays. We recommend reporting the effective number of patients with $K_i \geq 2$ alongside $hat\sigma_b^2$ and examining sensitivity to a_0, b_0.

4.3 Sensitivity Analysis for Informative Censoring

The frailty model and sensitivity analysis address complementatry aspects of informative censoring. Frailty b_i models heterogenity *parametrically*, assuming the Gaussian prior (14) and condition (17). The sensitivity analysis, on the other hand, *nonparametrically* tests the robustness of (17) without specifying a particular dependency structure. In practice, we fit the frailty logistic hazard then sweep the sensitivity parameter γ over a grid.

4.4 The Selection Model

The sensitivity analysis following (Scharfstein et al., 1999) and Bayesian treatent from Daniels and Hogan (2008). Define γ as the log-odds ratio of death at time $T_i = t$ between patients censored at t and those remaining under observation at t:

$$\gamma = \log\left(\frac{\Pr(T_i = t \mid T_i \geq t, C_i = t, \mathbf{x}_i, b_i)}{\Pr(T_i = t \mid T_i \geq t, C_i > t, \mathbf{x}_i, b_i)}\right) \tag{18}$$

$\gamma = 0$ corresponds to non-informative censoring (MAR), $\gamma < 0$ means censored patients have lower mortality risk (discharge reflects improvement), and $\gamma > 0$ means censored patients have higher mortality risk (discharge reflects deterioration). Here, γ plays the role of the selection parameter α in (8), indexing the residual selection bias beyond what b_i captures. This is because the frailty model only accounts for selection bias that can be captured by a patient-level random effect, but there may be further selection bias that is not captured by b_i such as time-varying information.

4.5 Tilted Likelihood

Under the selection model, the hazard for censored observations at its terminal interval is adjusted by the exponential tilting factor (analogous to α in (8)):

$$\text{logit}(\tilde{h}_{i,t}) = \text{logit}(h_{i,t}) + \gamma \mathbb{I}(C_i = t, \delta_i = 0), \qquad \text{where } \delta_i = \mathbb{I}(T_i \leq C_i) \tag{19}$$

Equivalently, the mortality odds are multiplied by $\exp(\gamma)$ at the censoring interval. For uncensored observations and non-terminal intervals, $\tilde{h}_{i,t} = h_{i,t}$. Then the tilted likelihood is:

$$\tilde{\mathcal{L}}(\boldsymbol{\theta}; \gamma) = \prod_{i=1}^{N} \prod_{t=1}^{K_i} \tilde{h}_{i,t}^{y_{i,t}} (1 - \tilde{h}_{i,t})^{1 - y_{i,t}} \tag{20}$$

The tilted likelihood is a reweighting of the original likelihood, where the contribution of censored observations at their terminal interval is adjusted by γ. This allows us to assess

how sensitive our inferences are to departures from the MAR assumption by varying γ and observing how the posterior distributions of parameters and survival curves change. The mortality odds are:

$$\frac{\tilde{h}_t}{1-\tilde{h}_t} = \frac{h_t}{1-h_t} \cdot \exp(\gamma) \quad \text{if } C_i = t, \delta_i = 0 \tag{21}$$

Since γ is fixed for each sweep and the censoring indicator $\mathbb{I}(C_i = t, \delta_i = 0)$ is observed, the tilt term $o_{i,t} = \gamma \mathbb{I}(C_i = t, \delta_i = 0)$ is a known offset for each person-period row. The tilted linear predictor is then:

$$\tilde{\eta}_{i,t} = \text{logit}(\tilde{h}_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + b_i + o_{i,t} \tag{22}$$

The offset term does not affect conjugate structure of PG sampler, only shifts the posterior mean.

4.6 Protocol.

Following Daniels and Hogan (2008):

1. Fit the frailty model with $\gamma = 0$ and record posterior summaries of $\boldsymbol{\beta}$, σ_b^2, and $\hat{S}(t \mid \mathbf{x})$.
2. Select a grid $\gamma \in \{-1.5, -1.0, -0.5, 0, 0.5, 1.0, 1.5\}$, where negative values correspond to the clinically plausible direction.
3. For each γ, refit using (20) with the PG sampler.
4. Report posterior survival curves and 95% credible intervals as a function of γ.
5. Identify the tipping point γ^* at which a substantive conclusion reverses sign.

If conclusions are robust across the grid of γ, the standard analysis assuming MAR is likely sufficient, despite the potential for informative censoring. If conclusions change at modest values of $|\gamma|$, this quantifies the fragility of the findings. A large γ^* (relative to the standard model) indicates that b_i has absorbed much of the informative censoring, leaving results less sensitive to residual violations. Conversely, a small γ^* after frailty adjustment suggests that the Gaussian random effect is insufficient to capture relative heterogeneity, pointing to potential misspecification or residual confounding.

5. Polya-Gamma Augmentation

6. Discussion

Appendix A. Semiparametric Estimation Equations under Informative Censoring

Reiterating a key result from Scharfstein et al. (1999) that pertains to this project, censoring induced confounding bias is a failure of the semiparametric estimation equations (score) for $\boldsymbol{\beta}$ when the MAR assumption is violated. When the probability of censoring depends on the unobserved outcome (the failure time T_i), this constitutes a Missing Not at Random (MNAR) or nonignorable censoring mechanism. In this case, the standard score functions used to estimate $\boldsymbol{\beta}$ are biased proportional to the dependence between T_i and C_i given $\mathbf{x}_i$.

Let $\Delta_i = \mathbb{I}(T_i \leq C_i)$ be the observation indicator (0 if censored, 1 if observed). A standard solution under MAR to account for missing data is to use an Inverse Probability Weighting (IPW). If MAR holds, the probability of being observed depends only on $\mathbf{x}_i$, denoted $\pi(\mathbf{x}_i; \alpha = 0) = \Pr(\Delta_i = 1 \mid \mathbf{x}_i)$. When MAR is violated, the true probability of being observed depends on both T_i and $\mathbf{x}_i$, denoted $\pi(T_i, \mathbf{x}_i; \alpha) = \Pr(\Delta_i = 1 \mid T_i, \mathbf{x}_i)$. Again, α is a non-identifiable selection parameter that indexes the severity of the MAR violation.

In standard parametric estimation, the expectation of the score at the true parameter value $\boldsymbol{\beta}_0$ always equals zero. Under informative censoring, however, this is no longer the case. To find the bias, we take the expectation under the MNAR distribution:

$$\text{Bias}(\alpha) = \mathbb{E}\left[\frac{\Delta_i}{\pi(\mathbf{x}_i; 0)} S_{\text{full}}(\boldsymbol{\beta}_0)\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{\Delta_i}{\pi(\mathbf{x}_i; 0)} S_{\text{full}}(\boldsymbol{\beta}_0) \middle| T_i, \mathbf{x}_i\right]\right] = \mathbb{E}\left[\frac{\pi(T_i, \mathbf{x}_i; \alpha)}{\pi(\mathbf{x}_i; 0)} S_{\text{full}}(\boldsymbol{\beta}_0)\right]$$

where $S_{\text{full}}(\boldsymbol{\beta})$ is the full-data score function (i.e., the score function we would have if we observed T_i for all patients). Following Scharfstein et al. (1999), if we specify the conditional hazard of censoring as $\lambda_C(t \mid \mathbf{x}_i, T_i) = \lambda_0(t \mid \mathbf{x}_i) \exp(\alpha T_i)$, where λ_0 is the baseline hazard of censoring, and $\exp(\alpha T_i)$ is the "tilt" caused by unobserved dependence between T_i and C_i. The survival function of censoring at T_i: $\pi(T_i, \mathbf{x}_i; \alpha) = \exp\{-\int_0^{T_i} \lambda_C(u \mid \mathbf{x}_i, T_i) du\} = \exp\{-\int_0^{T_i} \lambda_0(u \mid \mathbf{x}_i) \exp(\alpha T_i) du\} = \exp\{-\exp(\alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}$, where Λ_0 is the cumulative baseline hazard of censoring. While under MAR, $\pi(\mathbf{x}_i; 0) = \exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}$. Thus, the ratio of the true to assumed selection probabilities is:

$$\begin{aligned}
\frac{\pi(T_i, \mathbf{x}_i; \alpha)}{\pi(\mathbf{x}_i; 0)} &= \frac{\exp\{-\exp(\alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}}{\exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}} \\
&\approx \frac{\exp\{-(1 + \alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}}{\exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}} \\
&= \exp\{-\alpha T_i \Lambda_0(T_i \mid \mathbf{x}_i)\}
\end{aligned}$$

where $e^{\alpha T_i} \approx 1 + \alpha T_i$ in the tilt term on the numerator. Applying a first-order Taylor expansion for small α yields:

$$\frac{\pi(T_i, \mathbf{x}_i; \alpha)}{\pi(\mathbf{x}_i; 0)} \approx 1 - \alpha T_i \Lambda_0(T_i \mid \mathbf{x}_i) \tag{23}$$

Substituting this back into the expectation yields the explicit bias approximation:

$$\text{Bias}(\alpha) \approx -\alpha\, \mathbb{E}\Big[\Lambda_0(T_i \mid \mathbf{x}_i)\, T_i\, S_{\text{full}}(\boldsymbol{\beta})\Big] \tag{24}$$

This result explicitly demonstrates the main theoretical consequence of informative censoring: the bias in estimating $\boldsymbol{\beta}$ is strictly driven by α (the severity of the MAR violation/selection) and the covariance (cumulative hazard) between the failure time T_i and the censoring $\Lambda_0(T_i \mid \mathbf{x}_i)$. Because α cannot be identified from the observed data alone, standard estimation fails, motivating the need for alternative approaches such as sensitivity analyses or frailty models (as discussed in Section 4) to absorb this unobserved dependence. HOW TO CONNECT THIS TO THE FRAILTY MODEL/GLMM?

Appendix B. Derivation of Biased Hazard under Informative Censoring

At time t, the full population of survivors decomposes into those in the risk set $\mathcal{R}_t = \{i : T_i \geq t, C_i \geq t\}$ and those in $\mathcal{C}_t = \{i : T_i \geq t, C_i < t\}$ who have been censored by t, i.e. $\mathcal{S}_t = \mathcal{R}_t \cup \mathcal{C}_t$.

The true marginal hazard (h_t^*) and observed hazard ($\hat{h}_t$) are then:

$$
\begin{aligned}
h_t^* &= \Pr(T_i = t \mid T_i \geq t) = \mathbb{E}(h_{i,t} \mid T_i \geq t) \\
\hat{h}_t &= \Pr(T_i = t \mid T_i \geq t, C_i \geq t) = \mathbb{E}(h_{i,t} \mid T_i \geq t, C_i \geq t) = \mathbb{E}(h_{i,t} \mid i \in \mathcal{R}_t)
\end{aligned}
$$

Introduce $\lambda_t = \Pr(C_i < T_i \mid T_i \geq t) = |\mathcal{C}_t|/|\mathcal{R}_t|$ as the proportion of survivors (censored at t) relative to the risk set. Then, $\hat{h}_t$ can be expressed as a mixture of the hazards in $\mathcal{R}_t$ and $\mathcal{C}_t$. Let $h_t^c = \mathbb{E}(h_{i,t} \mid i \in \mathcal{C}_t)$ be the mean hazard among those censored by t. Then:

$$
\begin{aligned}
h_t^* &= (1-\lambda_t)\underbrace{\mathbb{E}(h_{i,t} \mid i \in \mathcal{R}_t)}_{\hat{h}_t} + \lambda_t \underbrace{\mathbb{E}(h_{i,t} \mid i \in \mathcal{C}_t)}_{h_t^c} \\
&= (1-\lambda_t)\hat{h}_t + \lambda_t h_t^c \\
\hat{h}_t &= \frac{h_t^* - \lambda_t h_t^c}{1-\lambda_t}
\end{aligned}
$$

Define $\Delta_t := h_t^* - h_t^c$ as the difference in mean hazards. Then:

$$
\begin{aligned}
\hat{h}_t &= \frac{h_t^* - \lambda_t(h_t^* - \Delta_t)}{1-\lambda_t} = \frac{h_t^* - \lambda_t h_t^* + \lambda_t \Delta_t}{1-\lambda_t} = h_t^* + \frac{\lambda_t \Delta_t}{1-\lambda_t} \\
\therefore \operatorname{Bias}(\hat{h}_t) &= \hat{h}_t - h_t^* = \frac{\lambda_t \Delta_t}{1-\lambda_t} > 0
\end{aligned}
$$

Whenever $\lambda_t, \Delta_t > 0$, the bias is positive, censoring rate is higher for healthier/sicker patients; $\lambda_t \to 1$ means more discharges implies risk set not representative (greater selection bias). For Δ_t greater differential among censored vs. risk set patients, greater selection bias, where this compounds in $\hat{S}_t = \prod_{s=1}^{t}(1-\hat{h}_s)$ so underestimation of S_t^* grows with t.

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